DATA STRUCTURE
LAB PROGRAM

12. Write a C program to implement the application of Stack (Notations)

```
#include <stdio.h>
#include <ctype.h>
#include <string.h>

#define MAX 100

char stack[MAX];
int top = -1;

void push(char ch) {
    stack[++top] = ch;
}

char pop() {
    return stack[top--];
}

char peek() {
    return stack[top];
}

int precedence(char op) {
    if (op == '^')
```

```
        return 3;
    if (op == '*' || op == '/')
        return 2;
    if (op == '+' || op == '-')
        return 1;
    return 0;
}

int main() {
    char infix[100], postfix[100];
    int i, j = 0;

    printf("Enter Infix Expression: ");
    scanf("%s", infix);

    for (i = 0; infix[i] != '\0'; i++) {
        char ch = infix[i];

        if (isalnum(ch)) {
            postfix[j++] = ch;
        }
        else if (ch == '(') {
            push(ch);
        }
        else if (ch == ')') {
            while (top != -1 && peek() != '(')
                postfix[j++] = pop();
            pop();
```

```
        }
        else {
            while (top != -1 && precedence(peek()) >= precedence(ch))
                postfix[j++] = pop();
            push(ch);
        }
    }

    while (top != -1)
        postfix[j++] = pop();

    postfix[j] = '\0';

    printf("Postfix Expression: %s\n", postfix);

    return 0;
}
```

Output

```
Enter Infix Expression: A+B*C
Postfix Expression: ABC*+


=== Code Execution Successful ===
```